Medical Classification using Bayesian Decision Theory

Assignment Report

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1 Introduction

In a medical system, the main job is to put patients into groups. Usually, we have two groups:

- ω_1: The patient is **Diseased**.
- ω_2: The patient is **Healthy**.

When a doctor does a test (like a blood test), they get a result which we call x. But the problem is that tests are not perfect. Sometimes a healthy person gets a high result, and a sick person gets a low result. This is why we need Bayesian Decision Theory. It helps us use math to make the best decision so that we don't make many mistakes.

2 Understanding the Math (Bayes' Rule)

To find out if a patient is sick, we need to find the "Posterior Probability." This is the chance that the patient is in a group *after* we see the test result.
The formula is:

$$P(\omega_i|x) = \frac{p(x|\omega_i)P(\omega_i)}{p(x)} \tag{1}$$

In simple words:

- $P(\omega_i|x)$ is what we want to find (Is the patient sick?).
- $p(x|\omega_i)$ is the Likelihood (What do test results usually look like for sick people?).
- $P(\omega_i)$ is the Prior (How many people in the world usually have this disease?).

3 How to Minimize Classification Error

The goal of this theory is to make the total number of wrong guesses as small as possible. This is called minimizing the "Classification Error."
To do this, we follow a very simple rule called the **Bayes Decision Rule**:

- If $P(\omega_1|x) > P(\omega_2|x)$, we say the patient is **Diseased**.
- If $P(\omega_2|x) > P(\omega_1|x)$, we say the patient is **Healthy**.

Why does this minimize error? Every time we make a choice, there is a chance we are wrong.

- If we pick ω_1, the chance of being wrong is $P(\omega_2|x)$.
- If we pick ω_2, the chance of being wrong is $P(\omega_1|x)$.

By always picking the one with the **higher** probability, we are automatically picking the one with the **lower** chance of error. If we do this for every patient, the total error of our medical system will be the lowest it can possibly be.

4 The Decision Boundary

Imagine a graph where the two classes overlap. The point where $P(\omega_1|x) = P(\omega_2|x)$ is called the **Decision Boundary**. On one side of this line, we classify everyone as sick. On the other side, we classify everyone as healthy. Bayesian theory helps us place this line at the exact spot where the overlapping "error area" is the smallest.

5 Medical Risk and Loss Functions

In a hospital, some mistakes are much worse than others. For example:

1. Telling a sick person they are healthy (This is very bad because they won't get medicine).

2. Telling a healthy person they are sick (This is bad but not as dangerous).

Bayesian theory uses something called a **Loss Function** (λ). It gives a "penalty score" to each mistake. Instead of just looking for the smallest error, we look for the **Minimum Risk**. We calculate it like this:

$$R(\alpha_i|x) = \sum_{j=1}^{2} \lambda(\alpha_i|\omega_j)P(\omega_j|x) \tag{2}$$

By choosing the class with the lowest Risk, the medical system becomes much safer for patients.

6 Conclusion

Bayesian Decision Theory is very helpful for medical systems. It uses old knowledge (priors) and new test results (likelihood) to make the best guess. It is the best way to make sure that the system makes the minimum number of mistakes and keeps the risk low for patients.

7 References

1. **Duda, R. O., Hart, P. E., & Stork, D. G.** *Pattern Classification.* John Wiley & Sons.

2. **Bishop, C. M.** *Pattern Recognition and Machine Learning.* Springer.

3. **Theodoridis, S., & Koutroumbas, K.** *Pattern Recognition.* Academic Press.